

ENTERPRISE NETWORK SECURITY ASSESSMENT AND DEFENSE PLANNING

Cybersecurity Project Report

Prepared By:

Lakshmi Subhaja Kari

Degree:

B.Tech – Cybersecurity

Project Role:

Junior Cyber Security Analyst

Assessment Environment:

Kali Linux + Metasploitable 2 + VMware Workstation

Project Type:

Authorized Defensive Security Assessment

Year:

2026

1. Executive Summary

This project presents an authorized network security assessment and defense planning exercise for a small-to-medium enterprise environment. The objective was to understand network architecture, identify exposed network services, analyze potential security threats, assess associated risks, and recommend appropriate defensive controls.

A controlled laboratory environment was created using Kali Linux as the security assessment workstation and Metasploitable 2 as the authorized target system. VMware Workstation was used to provide an isolated Host-only network.

Network discovery and service enumeration were performed using Nmap. The assessment identified multiple exposed services including FTP, Telnet, SSH, HTTP, SMB, database services, VNC, NFS, Tomcat and other network services.

The identified exposures were analyzed from a defensive perspective. Potential threats, impacts and recommended mitigations were documented in a risk assessment. Finally, a proposed enterprise network architecture was designed using firewall protection, network segmentation, IDS/IPS, secure wireless networking and access-control principles.

NO	OBJECTIVE
1.	Understand enterprise level architecture.
2.	Identify network vulnerabilities and misconfigurations.
3.	Analyse risks related to open ports and weak access controls.
4.	Analyze risks related to unsecured wireless networks.
5.	Design appropriate network design strategies.
6.	Develop a professional network security assessment

3. Scope and Scenario

The assessment is based on a hypothetical small-to-medium enterprise containing approximately 50–100 employees. The organization uses an internal LAN, Internet-facing services, employee Wi-Fi and guest Wi-Fi.

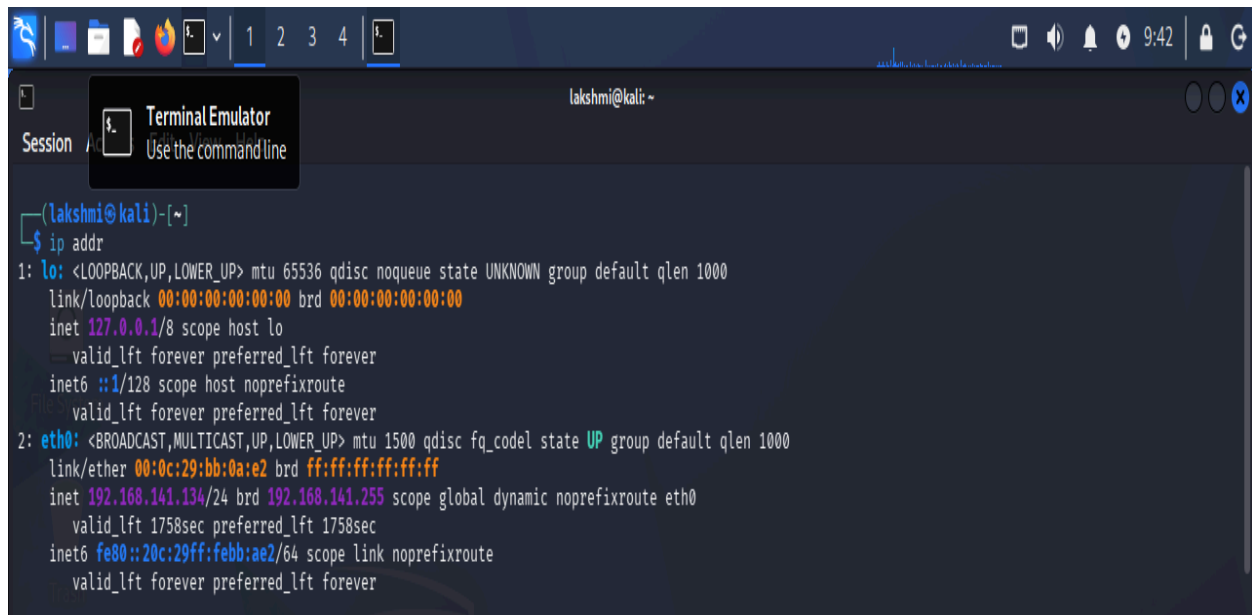
The scenario assumes that suspicious network behavior has been observed and that the organization requires a security assessment from a defensive perspective.

The assessment focuses on:

- Network architecture
- Network service exposure
- Potential unauthorized access
- Malware propagation
- Wireless security
- Network segmentation
- Firewall controls
- IDS/IPS monitoring
- Security risk assessment

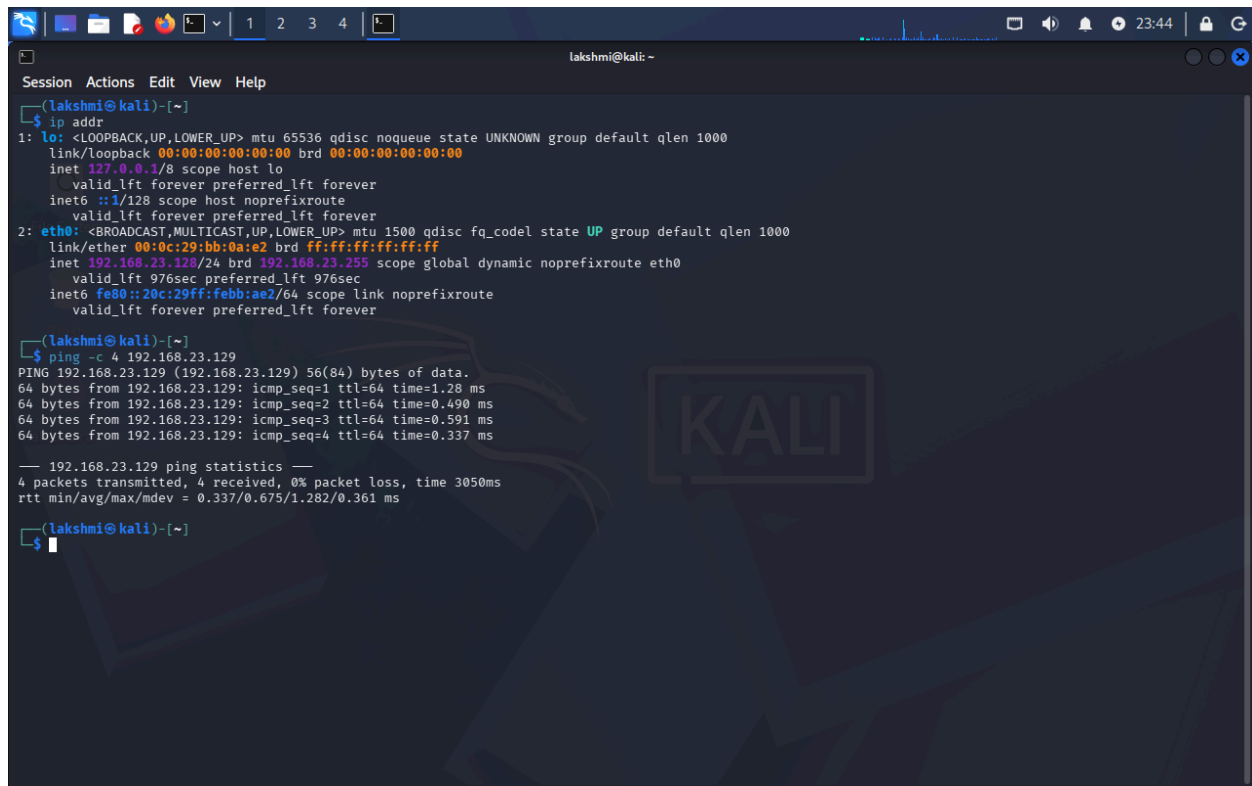
Testing was performed only against the authorized Metasploitable 2 lab laboratory system in an isolated VMware Host-only network.

COMPONENT	DETAILS
Security assessment machine kali linux	Kali Linux
Authorized Target	Metasploitable 2
Virtuvalization	VMware Workstation
Network Type	Host only
Target IP	192.168.23.129
Network	192.168.23.0/24
Primary Tool	NMAP
Assesement Type	Authorized laboratory assesement



```
(lakshmi@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bb:0a:e2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.141.134/24 brd 192.168.141.255 scope global dynamic noprefixroute eth0
        valid_lft 1758sec preferred_lft 1758sec
    inet6 fe80::20c:29ff:febb:ae2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

FIG:1 I performed the ip addr for my Kali linux.



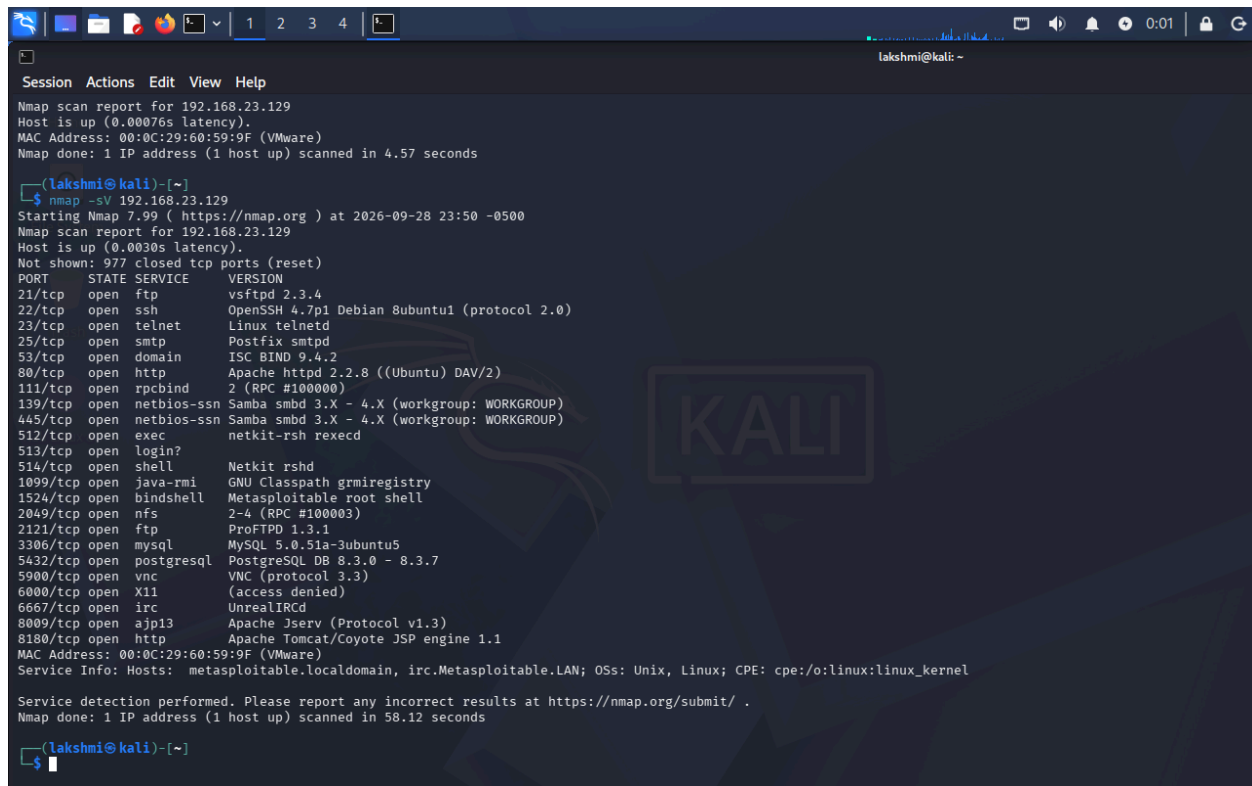
```
(lakshmi@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bb:0a:e2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.128/24 brd 192.168.23.255 scope global dynamic noprefixroute eth0
        valid_lft 976sec preferred_lft 976sec
    inet6 fe80::20c:29ff:febb:ae2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(lakshmi@kali)-[~]
$ ping -c 4 192.168.23.129
PING 192.168.23.129 (192.168.23.129) 56(84) bytes of data.
64 bytes from 192.168.23.129: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 192.168.23.129: icmp_seq=2 ttl=64 time=0.490 ms
64 bytes from 192.168.23.129: icmp_seq=3 ttl=64 time=0.591 ms
64 bytes from 192.168.23.129: icmp_seq=4 ttl=64 time=0.337 ms

--- 192.168.23.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3050ms
rtt min/avg/max/mdev = 0.337/0.675/1.282/0.361 ms

(lakshmi@kali)-[~]
$
```

FIG:2 I verified network communication between the Kali Linux assessment machine and the Metasploitable 2 target through the VMware Host-only network.”



```
Session Actions Edit View Help
Nmap scan report for 192.168.23.129
Host is up (0.00076s latency).
MAC Address: 00:0C:29:60:59:9F (VMware)
Nmap done: 1 IP address (1 host up) scanned in 4.57 seconds

(lakshmi@kali)-[~]
$ nmap -sn 192.168.23.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-28 23:50 -0500
Nmap scan report for 192.168.23.129
Host is up (0.0030s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:60:59:9F (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 58.12 seconds

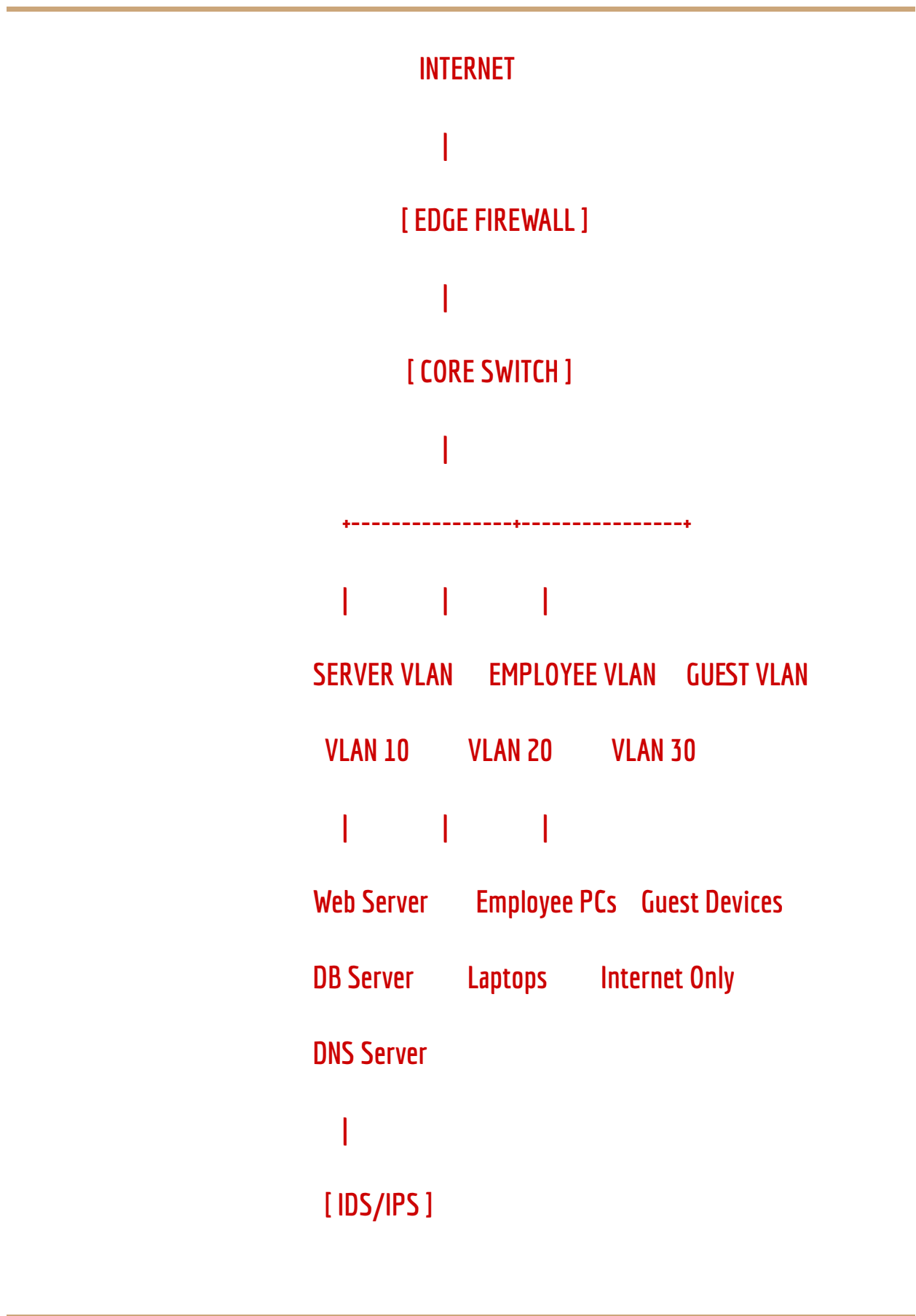
(lakshmi@kali)-[~]
$
```

FIG :3 I performed host discovery using Nmap with the command `nmap -sn 192.168.23.129`. This scan was used to check whether the host with IP address 192.168.23.129 was active and reachable on the network. The `-sn` option performs host discovery without scanning the ports. The scan successfully identified the target host as up, confirming that the device was reachable and responding on the network.

PORT	SERVICE
21	FTP
22	SSH
23	TELNET
25	SMTP
53	DNS
80	HTTP
139/445	SMB
3306	MYSQL
5432	POSTGRESQL
5900	VNC
8009	AJP
8180	HTTP

THREAT	AFFECTED AREA	POTENTIAL IMPACT	CONTROL
Unauthorized remote access	Telnet	Unauthorized login attempts	Disable Telnet Use SSH
Unautrhorized file access	FTP	Data exposure	Use SFTP / Restrict FTP
Malware propagation	SMB	Malware spreading	Network Segmentation
Database Compromise	MYSQL/POSTGRESQ L	Data exposure	Restrict database access
Wireless attacks	WIFI	Unauthorized Network access	WPA2/WPA3+guest isolation

FINDING	EVIDENCE	LIKELIHOOD	IMPACT	RISK
Telnet exposed	TCP 23	HIGH	HIGH	HIGH
SMB exposed	TCP 139/445	HIGH	HIGH	HIGH
MYsql exposed	TCP 3306	MEDIUM	HIGH	HIGH
VNC exposed	TCP 5900	MEDIUM	HIGH	HIGH
HTTP Exposed	TCP 80	MEDIUM	MEDIUM	MEDIUM



RULE	SOURCE	DESTINATION	ACTION
1.	INTERNET	INTERNAL NETWORK	DENY UNNECESSARY TRAFFIC
2.	EMPLOYEE VLAN	INTERNET	ALLOW REQUIRED TRAFFIC
3.	GUEST VLAN	INTERNAL VLANS	DENY
4.	ADMIN VLAN	SERVERS	ALLOW AUTHORIZED MANAGEMENT
5.	EMPLOYEE VLAN	DATABASE	DENY UNLESS REQUIRED

12. Conclusion

The Enterprise Network Security Assessment and Defense Planning project provided practical experience in network discovery, service enumeration, threat identification, risk assessment and defensive security planning.

The authorized laboratory assessment identified multiple exposed network services on the Metasploitable 2 system. These exposures were analyzed as potential security risks rather than as evidence of successful compromise.

The assessment demonstrated the importance of minimizing unnecessary network services, restricting access to sensitive systems, implementing network segmentation, securing wireless networks, monitoring network activity and maintaining appropriate security controls.

The proposed defense architecture combines firewall protection, VLAN-based segmentation, IDS/IPS monitoring, secure wireless configuration and access-control principles to reduce the organization's attack surface and improve its overall security posture.